

Article number

5SY6106-7

Miniature circuit breaker 230/400 V 6kA, 1-pole, C, 6 A, D=70 mm



Model

product brand name	SENTRON
product designation	Miniature circuit breaker

General technical data

number of poles	1
design of pole	1P
tripping characteristic class	C
mechanical service life (operating cycles) typical	10 000
overvoltage category	3
degree of pollution	3

Voltage

type of voltage of the operating voltage	AC
insulation voltage (Ui)	
• with single-phase operation at AC rated value	440 V
• with multi-phase operation at AC rated value	440 V
supply voltage with single-phase operation at AC rated value	230 V
• operational current	
— at 40 °C rated value	5.63 A
— at 50 °C rated value	5.24 A
— at 55 °C rated value	5.04 A
• operational current at AC rated value	6 A

Supply voltage

supply voltage	
• at AC	400 V
• at DC rated value	60 V
value range of the supply voltage frequency	50/60 Hz
operating voltage at DC rated value maximum	72 V

Protection class

protection class IP	IP20, with connected conductors
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Breaking Capacity

switching capacity current	
• at DC according to IEC 60947-2 rated value	15 kA
• according to EN 60898 rated value	6 kA
• according to IEC 60947-2 rated value	30 kA
energy limitation class	3

Dissipation

power loss [W] for rated value of the current at AC in hot operating state per pole	1.6 W
suitability for operation	Infrastructure / Industry

Product details

product feature touch protection	Yes
product component	

● combined terminal top	Yes
● combined terminal bottom	Yes
● neutral conductor switching	No
product feature properties for main switches in accordance with EN 60204-1	Yes
product feature halogen-free	Yes
product feature sealable	Yes
product feature silicon-free	Yes
product extension installable supplementary devices	Yes

Product function

set values setting current (Ii) for I-tripping	8
reference value setting current (Ii) for I-tripping	x I _n

Short circuit

short-circuit current breaking capacity (I_{cn})	
● at AC according to UL 1077 and CSA C22.2 No.235	5 kA

Connections

connectable conductor cross-section solid	
● minimum	0.75 mm ²
● maximum	35 mm ²
connectable conductor cross-section stranded	
● minimum	0.75 mm ²
● maximum	35 mm ²
connectable conductor cross-section finely stranded with core end processing	
● minimum	0.75 mm ²
● maximum	25 mm ²
AWG number as coded connectable conductor cross section	
● minimum	18
● maximum	4
tightening torque [lbf·in] with screw-type terminals	
● minimum	22 lbf·in
● maximum	31 lbf·in
tightening torque with screw-type terminals	
● minimum	2.5 N·m
● maximum	3.5 N·m
position of power supply cord	Any

Mechanical Design

height	90 mm
width	18 mm
depth	76 mm
installation depth	70 mm
number of modular width units	1
fastening method	DIN rail
mounting position	any
Net Weight	150.1 g

Environmental conditions

influence of the surrounding temperature	max. 95% RH up to 55°C
standard	IEC / EN 60898-1, IEC / EN 60947-2 / UL1077
vibration resistance according to IEC 60068-2-6	±1mm at 5 to 25Hz; 50m/s ² at 25 to 150Hz
ambient temperature during operation	
● minimum	-25 °C
● maximum	55 °C
ambient temperature during storage	
● minimum	-40 °C
● maximum	75 °C

number of test cycles for environmental testing according to IEC 60068-2-30 6

Approvals Certificates

Environment

Environmental Confirmations [Yes](#)

EPD Typ II/III (with life cycle assessment)  EPD Typ II/III (with life cycle assessment) [Yes](#)

Environmental Confirmations [Yes](#)

General Product Approval

CCC  CCC [Yes](#)

IMQ  IMQ [Yes](#)

cURus  cURus [Yes](#)

VDE  VDE [Yes](#)

Miscellaneous [Yes](#)

KC  KC [Yes](#)

EAC  EAC [Yes](#)

S Mark  S Mark [Yes](#)

Confirmation [Yes](#)

EU declaration of conformity  EU declaration of conformity [Yes](#)

UK Declaration of Conformity  UK Declaration of Conformity [Yes](#)

EMV

RCM  RCM [Yes](#)

Test Certificates

Miscellaneous [Yes](#)

Maritime application

ABS (American Bureau of Shipping)  ABS (American Bureau of Shipping) [Yes](#)

BV (Bureau Veritas)  BV (Bureau Veritas) [Yes](#)

DNV (Det Norske Veritas)  DNV (Det Norske Veritas) [Yes](#)

LR (Lloyds Register)  LR (Lloyds Register) [Yes](#)

RINA (Registro Italiano Navale)  RINA (Registro Italiano Navale) [Yes](#)

other

Chinese Quality Checkmark  Chinese Quality Checkmark [Yes](#)

Confirmation [Yes](#)

Further information

Information on the packaging

<https://support.industry.siemens.com/cs/ww/en/view/109813875>

Information for data generation and storage

<https://support.industry.siemens.com/cs/ww/en/view/109995012>

Information- and Downloadcenter (Catalogs, Brochures,...)

<https://www.siemens.com/lowvoltage/catalogs>

Industry Mall (Online ordering system)

<https://mall.industry.siemens.com/mall/en/en/Catalog/product?mlfb=5SY6106-7>

Service&Support (Manuals, Certificates, Characteristics, FAQs,...)

<https://support.industry.siemens.com/cs/ww/en/ps/5SY6106-7>

Image database (product images, 2D dimension drawings, 3D models, device circuit diagrams, ...)

https://www.automation.siemens.com/bilddb/cax_en.aspx?mlfb=5SY6106-7

CAX-Online-Generator

<https://www.siemens.com/cax>

Tender specifications

<https://www.siemens.com/specifications>

Characteristic curves

[https://curves.simaris.siemens.com/curves/<mmp_prod_noCOMP="HAUPT"></mmp_prod_no>](https://curves.simaris.siemens.com/curves/<mmp_prod_noCOMP=)

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